

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Bijoy Varghese	Reg. No.:	192411013
Section / Batch:	2024 batch (3 rd yr)	Date:	22/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Code of Conduct

I Bijoy Varghese (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bijoy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1	4	4	3	4	3	46
Q2	4	4	3	3	4	45
Overall Total (100)						91

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1) Concept Map: Algorithm $\rightarrow$ Time Complexity $\rightarrow$ Asymptotic Notation $\rightarrow$ (Big O, Big-Theta, Big Omega)

ALGORITHM
(step-by-step procedure to solve a problem.)

Input Size(n)
(Size of the data.)

Time Complexity.
(Growth of operation with respect to n.)

Analysis Perspective

Best Case

Average Case

Worst Case

Asymptotic Notation
(Mathematical way to describe growth for large n.)

Best case Analysis

Average Case Analysis

Worst case Analysis

Big - O
 $O(g(n))$

Big - THETA
 $\Theta(g(n))$

Big - OMEGA
 $\Omega(g(n))$

GROWTH RATE
(How the fn behaves as $n \rightarrow \infty$)

Efficiency & Scalability
(Determine Performance of algorithm)

Q2) Concept Map - Divide & Conquer $\rightarrow$ Recurrence Relation
 $\rightarrow$ Master Theorem $\rightarrow$ Time Complexity.

Divide & Conquer.
(Algorithm Paradigm)

RECURRENCE RELATION
(Represents running time)
 $T(n) = aT(n/b) + f(n)$

a
No. of subproblems

b
Factor by
which problem
size is
reduced.

$f(n)$
Work done
outside
recursive
calls.

Recursion
Tree.
(Alternative
Analysis.)

MASTER'S THEOREM
Compare $f(n)$ with $n^{\log_b a}$

Case 1
If $f(n) = O(n^{\log_b a - \epsilon})$
for some $\epsilon > 0$

$$T(n) = \Theta(n^{\log_b a})$$

Case 2
If $f(n) = \Theta(n^{\log_b a} \log^k n)$
for $k \geq 0$

$$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

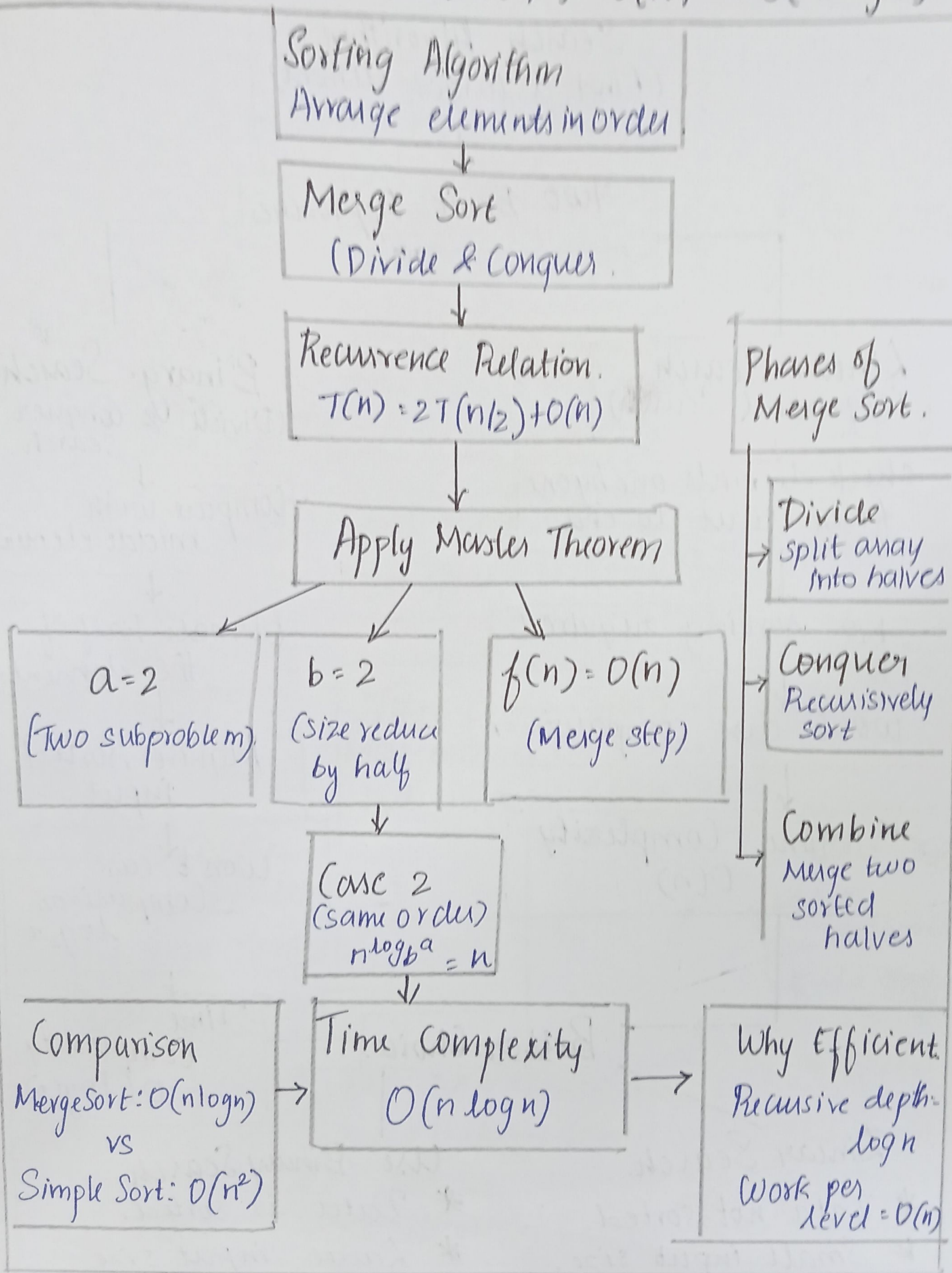
Case 3.
If $f(n) = \Omega(n^{\log_b a + \epsilon})$
for some $\epsilon > 0$

$$T(n) = \Theta(f(n))$$

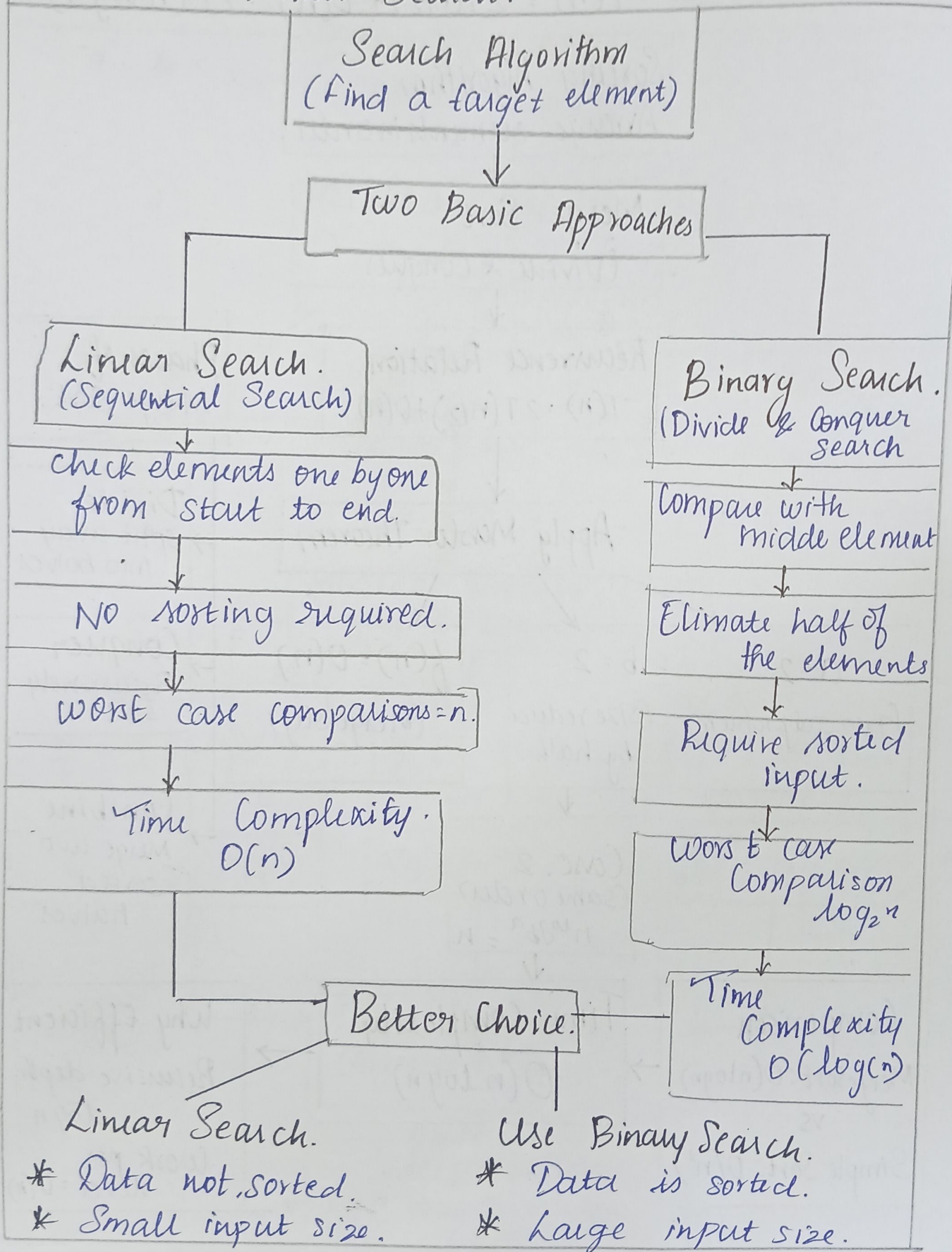
Time Complexity.
(Asymptotic Running time)

Interpretation.
Gives growth rate of
algorithm

Q5) Concept Map: Sorting Algorithm $\rightarrow$ Merge Sort $\rightarrow$
Recurrence $T(n) = 2T(n/2) + O(n) \rightarrow O(n \log n)$



Q6) Concept Map: Search Algorithm $\rightarrow$ Binary Search vs Linear Search.



Q8) Concept Map: Algorithm $\rightarrow$ Space Complexity $\rightarrow$ Efficiency Trade-off Decision.

